

Holographic Parallax

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Figure 1: Our holographic display prototype and experimentally captured 4D light field holograms. (*left*) The prototype utilizes time-multiplexing to produce speckleless, high-contrast 4D light field holograms. The captured images of 4D light field holograms exhibit (*middle*) accurate focus cues with natural blurs and (*right*) parallax cues corresponding to the dynamic viewpoint. (Low Poly Series credits to Stoolfeather Games, Dragon and Bunny credit to Stanford Computer Graphics Laboratory.)

ABSTRACT

Cutting-edge computer-generated holography (CGH) algorithms have realized significant advancements in reconstructing the desired light field, enabling the creation of 4D light field holograms. These holograms offer view-dependent effects, such as motion parallax, occlusion, and specular reflections, which are crucial for enhancing the perceptual realism of 3D scenes. Our holographic

display prototype leverages sophisticated CGH algorithms in conjunction with a high-speed binary spatial light modulator (SLM) to deliver an immersive experience of 4D light field holograms. This demonstration highlights the latest progress in CGH technology and emphasizes the critical role of parallax cues in providing perceptually realistic 3D contents.

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1 INTRODUCTION

Holographic displays have long been recognized as next-generation 3D displays for virtual reality (VR) and augmented reality (AR).

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Table 1: Comparison of recent Emerging Technologies demos in 3D near-eye displays.

	retinal blur	occlusion boundary	view dependency
Single-Shot VR [Qin et al. 2023]	correct	no	no
Neural Holography [Choi et al. 2023]	correct	approx.	approx.
Holographic Parallax (Ours)	correct	correct	correct

Owing to their flexibility in wavefront modulation, holographic displays can provide 3D depth cues, enhancing perceptual realism. Recent advancements in computer-generated holography (CGH) have underscored its potential in providing real-time high-quality 3D holograms through artificial intelligence-based methods. Through iterative optimization or neural networks, holograms are generated to reconstruct specific target contents. Generally, CGH algorithms utilize various formats of target 3D contents, such as multiple image layers, RGB-depth images, or focal stacks.

While most 3D data formats represent the 3D scene from a single central view and focus on providing accommodation cues [Kim et al. 2022], the recently introduced 4D light field holograms have showcased the ability of holographic displays to reconstruct 3D scenes with view-dependent visual effects. Considering that the motion parallax serves as the most strongest depth cue for objects one meter or more away, 4D light field holograms hold the potential to significantly enhance the perceptual realism of 3D scenes. [Kim et al. 2024] In this work, we present an unprecedented quality of 4D light field hologram experience, leveraging state-of-the-art CGH optimization methods combined with time-multiplexing techniques [Choi et al. 2022]. The presence of view-dependent visual effects, such as parallax, occlusion, and specular reflection, improves the perceptual realism of 3D scenes in near-eye display.

2 METHODS

CGH algorithms calculate phase or amplitude patterns of the spatial light modulator (SLM), which reconstruct wavefront of the desired 3D scene. In this work, we apply CGH optimization method developed by Choi et al. [2022], which utilizes the short-time Fourier transform and stochastic gradient descent to optimize the SLM phase patterns for 4D light field data. Since 4D light field holograms necessitates the time-multiplexing of multiple frames of holograms to reduce speckle noise and address the over-constrained problem of CGH optimization, we employ a high-speed binary SLM (QXGA-R10, Forth Dimension Display). Please refer to our technical paper [Kim et al. 2024] for implementation details. The left column of Fig. 1 features our monocular near-eye display prototype. A beam splitter is placed before the eyepiece, allowing one beam path to be captured live during the demonstration, which is then monitored by the exhibitor to guarantee safety.

3 DEMO EXPERIENCE

Participants will have the opportunity to explore 4D light field holograms through our monocular holographic near-eye display prototype. The participants will be guided to sit in front of the display and place their head on a chin rest. A short sequence of holograms (less than 3 minutes) will be presented, during which participants are encouraged to actively engage with the holograms by moving their eyes or head. Initially, same scene with various

target data formats (2D plane, 3D focal stack, and 4D light field) will be presented. This will allow participants to observe the differences in focus cues and parallax cues across different 3D data formats. Subsequently, we will showcase multiple scenes in 4D light field holograms to offer a diverse experience of light field scene perception. This demonstrates the enhanced perceptual realism offered by 4D light field technology, particularly through parallax cues.

To the best of our knowledge, this work presents the first demonstration of 4D light field holograms capable of supporting view-dependent effects. It is worth noting that multifocal displays [Qin et al. 2023] and 3D holographic displays [Choi et al. 2023] have been showcased at Emerging Technologies in 2023. While these works have successfully provided focus cues, they either neglected or approximated the occlusion boundary or view-dependent visual effects (See Table 1). Our 4D light field hologram provides a high-resolution, dense light field to a single eyepiece, enabling various visual cues. Furthermore, the use of a light field target with a binary SLM ensures a uniform energy distribution across the eyepiece, effectively reducing visual artifacts associated with eye floaters.

4 CONCLUSION

Our work provides viewing experience of 4D light field holograms that accurately reconstruct the light field of the target 3D scene, highlighting the recent advancements in computational holography and offering valuable insights for the development of perceptually realistic 3D displays.

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